

Kristopher Manalo
<https://github.com/KristopherManalo>

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EDUCATION

University of California, San Diego

B.S. Computer Engineering | June 2026

Relevant Coursework: Databases, Networked Services, Computer Security

SKILLS

Languages: Python, C, C++, Rust, Go, Java, SystemVerilog, ARM Assembly, x86 Assembly, Bash

Databases & Tools: PostgreSQL, Git, UNIX/Linux Terminals, Intel Quartus Prime

Other: American Sign Language (biliterate)

EXPERIENCE

Web Engineering Intern | KSDT Radio @UCSD | October 2025-April 2026

- Developed features for a web application serving content to radio station listeners
- Collaborated with cross-functional teams to define requirements, milestones, and feature scope
- Translated product requirements into implemented features, contributing to application functionality and user experience

PROJECTS

Custom RISC Architecture and Toolchain (SystemVerilog + Assembler)

- Designed an Instruction Set Architecture for a custom RISC 8-bit machine, defining instructions, encodings, and execution semantics.
- Implemented a synthesizable single-cycle CPU in SystemVerilog and validated functionality by synthesizing with Intel Quartus Prime
- Built an assembler with code generation techniques to abstract irregular instruction encodings, improving maintainability and ease of use
- Executed end-to-end programs on CPU, verifying correct instruction execution and overall system behavior

Android Reminder App (Agile Team Project)

- Developed user-facing features for reminder-based Android app over a 10-week agile period with 2 main milestones
- Coordinated with team members, to define goals, break down tasks, and track progress through sprint-based workflows
- Delivered functioning app features meeting milestone objectives

Ray Tracing Renderer (C++ / OpenGL)

- Implemented a ray tracing renderer with OpenGL for GPU interaction, supporting a user-defined number of rays and bounce depth.
- Modeled Scene objects with transparency, reflectivity, and material properties.